Chinmay Shah

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EDUCATION

Institute of Human Machine and Cognition (University of West Florida) — ***Aug 2023 - Present***
PhD in Intelligent Systems and Robotics — *(GPA 4.0/4.0)*

North Carolina State University — ***Aug 2019 – Dec 2021***
MS(Thesis) in Mechanical Engineering — *(GPA 4.0/4.0)*

Vishwakarma Institute of Technology (VIT) — ***Aug 2014 – May 2018***
B.Tech. in Mechanical Engineering — *(GPA 9.12/10.0)*

PUBLICATIONS

- C. Shah, J. Li, G. Clark, "**Real-Time Phase Conditioned Human Motion Forecasting For Wearable Robot Control**" submitted to IROS, Pittsburg, 2026
- C. Shah, A. Fleming, V. Nalam, M. Liu and H. H. Huang, "**EMG-driven Musculoskeletal model for volitional control of a robotic ankle prosthesis**" presented at IROS, Kyoto, 2022 https://ieeexplore.ieee.org/document/9981305
- S. Upadhye, C. Shah, M. Liu, G. Buckner, and H. H. Huang, "**A Powered Prosthetic Ankle Designed for Task Variability - A Concept Validation**," presented at the International Conference on Intelligent Robots and Systems (IROS), Prague, 2021. https://ieeexplore.ieee.org/document/9636324

RESEARCH & PROFESSIONAL EXPERIENCE

Institute of Human Machine and Cognition (IHMC) — ***Aug 2023 - Present***

Research Assistant

PhD Thesis: *Intent-Aware Representations of Human Motion For Proactive Exoskeleton Assistance*

PhD Advisor: *Dr. Geoffery Clark and Dr. Greg Sawicki*

- Trained a **phase-based periodic autoencoder** for human motion prediction on wearable data, achieving strong generalization across **unseen users and locomotion modes (MAE 4–8°, R^2 > 0.75)**.
- Developing a **Fourier Latent Dynamics (FLD)** framework for exoskeleton control, learning compact and interpretable representations (e.g., phase, frequency) from wearable sensors to predict user intent and **enable adaptive assistance across users and gait conditions**.
- Designed and planning to collect a **multi-subject locomotion dataset** (10 participants) containing motion, EMG and pressure insole data; covering walking, stairs, ramps, turns, and speed transitions.
- Developed end-to-end pipelines for sensor synchronization, processing motion capture data and biomechanics inverse kinematics and kinetics analysis.
- Applied a reduced order inverted pendulum model to **estimate shear forces**, improving gravity compensation and stability during exoskeleton-assisted walking, pushing, pulling.

Wearable Sensor Suit Development

- Designed a **wearable sensor suit** with 7 body-mounted IMUs and 2 pressure insoles, decoupling sensing from actuation to capture unbiased user intent; streams synchronized multi-modal data at 200 Hz for offline training and real-time onboard inference.
- Deployed **Temporal Convolutional Network (TCN)** models for real-time inverse kinematics (IK) and inverse dynamics (ID) estimation, **enabling portable, lab-free biomechanics assessment**.
- Currently developing custom **wireless IMU modules** to eliminate wiring constraints, extending the platform toward a full-body suit or modular joint specific system for **continuous health monitoring**.
- Working toward leveraging the suit as a foundation for subject-specific **biomechanical digital twins**, with the goal of **rehabilitation monitoring**, and **early assessment of neurological and musculoskeletal impairments.**

DEKA Research and Development ***Jan 2022 – Aug 2023***
Robotics Control Systems Engineer

- Contributed to the path planning stack for an autonomous last-mile delivery robot (FedEx) and an autonomous security robot, taking systems from concept toward market-ready deployment.
- Developed the high-level **behavior planner** coordinating global path generation and context-specific controllers for sidewalk, road, and intersection navigation, including pedestrian yielding; implemented core path planning algorithms (A*, RRT, Grassfire).
- Engineered a **hierarchical collision-check library** leveraging ML-based trajectory predictions for dynamic obstacle avoidance, reducing collision incidents by ~16% in real-world testing.
- Introduced **reachability and Euclidean-distance cost layers** into the Model Predictive Controller (MPC) for safe navigation in narrow indoor environments; leveraged GPU programming to maintain real-time performance.

Neuromuscular Rehabilitation Engineering Lab ***Mar 2020 – Dec 2021***

Research Assistant

- Developed an **EMG-driven musculoskeletal model controller** for a robotic ankle prosthesis, replicating the biological neuromuscular pathway by mapping user EMG signals through a hill-type muscle model to generate volitional torque commands — effectively closing the neural-to-mechanical loop severed by amputation.
- Designed a **2-muscle lumped-parameter Hill-type model** and implemented a non-linear optimization scheme in MATLAB for subject-specific parameter tuning, achieving >85% torque prediction accuracy; validated the model through inverse dynamics and kinematics analyses.
- Implemented **Least Squares Policy Iteration (LSPI)** for adaptive gait assistance on a hip exoskeleton, reducing metabolic cost while preserving walking efficiency — extending the biologically-grounded control philosophy from prosthesis to assistive exoskeleton.

TECHNICAL EXPERTISE

Software: MUJOCO, Isaac Sim, OpenSim, ROS, Git, MATLAB/Simulink, TwinCAT 3.1 (EtherCAT), SolidWorks, ANSYS, VICON, Nimble Physics

Programming Languages: C, C++, Python, Java

Python Packages: PyTorch, NumPy, SciPy, scikit-learn, pandas, Matplotlib, Plotly, TensorBoard, OpenCV, Gymnasium, Stable-Baselines3, SQLite